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Q-1	Write a program to declare all primitive data types (`byte`, `short`, `int`, `long`, `float`, `double`, `char`, `boolean`) and print their default values and memory size using `Byte.SIZE`, `Integer.SIZE`, etc.

Code	class Pro1
	{
	byte a;
	short b;
	int c;
	long d;
	float e;
	double f;
	char g;
	boolean h;
	public static void main(String[] args)
	{
	Pro1 p1=new Pro1();
	System.out.println("==========");
	System.out.println("\tDefault value\t||\tSize(Bits)\t||\tSize(Bytes)");
	System.out.println("==========");
	System.out.println("byte\t\t"+p1.a+"\t||\t"+Byte.SIZE+"\t\t||\t"+(Byte.SIZE)/8);
	System.out.println("short\t\t"+p1.b+"\t||\t"+Short.SIZE+"\t\t||\t"+(Short.SIZE)/8);
	System.out.println("int\t\t"+p1.c+"\t||\t"+Integer.SIZE+"\t\t||\t"+(Integer.SIZE)/8);
	System.out.println("long\t\t"+p1.d+"\t||\t"+Long.SIZE+"\t\t||\t"+(Long.SIZE)/8);
	System.out.println("float\t\t"+p1.e+"\t||\t"+Float.SIZE+"\t\t||\t"+(Float.SIZE)/8);
	System.out.println("double\t\t"+p1.f+"\t||\t"+Double.SIZE+"\t\t||\t"+(Double.SIZE)/8);
	System.out.println("char\t\t"+"NULL"+"\t||\t"+Character.SIZE+"\t\t||\t"+(Character.SIZE)/8);
	System.out.println("boolean\t\t"+p1.h+"\t||\tNot a (.SIZE)\t||\tDepend on JVM");
	}
	}

Output	E:\RK_P\java\tutorials\t_2>javac Pro1.java E:\RK_P\java\tutorials\t_2>java Pro1 ========== Default value || Size(Bits) || Size(Bytes) ========== byte 0 || 8 || 1 short 0 || 16 || 2 int 0 || 32 || 4 long 0 || 64 || 8 float 0.0 || 32 || 4 double 0.0 || 64 || 8 char NULL || 16 || 2 boolean false || Not a (.SIZE) || Depend on JVM
Q-2	Write a program to demonstrate **implicit type casting** (widening) and



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	explicit type casting (narrowing) — e.g., `int` to `double` and `double` to `int`, and show data loss when narrowing.

Code	class Pro2
	{
	public static void main(String[] args)
	{
	byte a=23;
	short b=56;
	int c=123;
	long d=987;
	float e=245.87f;
	double f=987.78;
	char g='A';
	boolean h=true;
	
	System.out.println("");
	System.out.println("\tImplicit type casting(widening)");
	
	System.out.println("==========
	==========");
	System.out.print("short\t<= byte :\t");
	b=a;
	System.out.println(b);
	System.out.print("int\t<= byte :\t");
	c=a;
	System.out.println(c);
	System.out.print("long\t<= byte :\t");
	d=a;
	System.out.println(d);
	System.out.print("float\t<= byte :\t");
	e=a;
	System.out.println(e);
	System.out.print("double\t<= byte :\t");
	f=a;
	System.out.println(f);
	
	System.out.println("");
	System.out.println("");
	
	b=76;
	System.out.print("int\t<= short :\t");
	c=b;
	System.out.println(c);
	System.out.print("long\t<= short :\t");
	d=b;
	System.out.println(d);
	System.out.print("float\t<= short :\t");



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```
e=b;
System.out.println(e);
System.out.print("double\t<=  short :\t");
f=b;
System.out.println(f);

System.out.println("");
System.out.println("");
System.out.println("ASCII VALUE ");
System.out.print("int\t<=  char :\t");
c=g;
System.out.println(c);
System.out.print("long\t<=  char :\t");
d=g;
System.out.println(d);
System.out.print("float\t<=  char :\t");
e=g;
System.out.println(e);
System.out.print("double\t<=  char :\t");
f=g;
System.out.println(f);

System.out.println("");
System.out.println("");
c=567;
System.out.print("long\t<=  int :\t");
d=c;
System.out.println(d);
System.out.print("float\t<=  int :\t");
e=c;
System.out.println(e);
System.out.print("double\t<=  int :\t");
f=c;
System.out.println(f);


System.out.println("");
System.out.println("");
d=9876;

System.out.print("float\t<=  long :\t");
e=d;
System.out.println(e);
System.out.print("double\t<=  long :\t");
f=d;
System.out.println(f);

System.out.println("");
```



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```
                    System.out.println("");
                    e=89565.99f;
                    System.out.print("double\t<=  float :\t");
                    f=e;
                    System.out.println(f);



                    System.out.println("");
                    System.out.println("\tExplicit type casting(narrowing)");
System.out.println("==========================================================
============");
                    f=9876.98;
                    System.out.print("byte\t<= double :\t");
                    a=(byte)f;
                    System.out.println(a);
                    System.out.print("short\t<=  double :\t");
                    b=(short)f;
                    System.out.println(b);
                    System.out.print("int\t<=  double :\t");
                    c=(int)f;
                    System.out.println(c);
                    System.out.print("long\t<=  double :\t");
                    d=(long)f;
                    System.out.println(d);
                    System.out.print("float\t<=  double :\t");
                    e=(float)d;
                    System.out.println(e);
                    System.out.print("char\t<=  double :\t");
                    g=(char)f;
                    System.out.println(g);

                    System.out.println("");
                    System.out.println("");

                    e=99999.876f;
                    System.out.print("byte\t<=  float :\t");
                    a=(byte)e;
                    System.out.println(a);
                    System.out.print("short\t<=  float :\t");
                    b=(short)e;
                    System.out.println(b);
                    System.out.print("int\t<=  float :\t");
                    c=(int)e;
                    System.out.println(c);
                    System.out.print("long\t<=  float:\t");
                    d=(long)e;
                    System.out.println(d);
```



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```
System.out.print("char\t<=  float :\t");
g=(char)e;
System.out.println(g);

System.out.println("");
System.out.println("");
d=85;
System.out.print("byte\t<=  long :\t");
a=(byte)d;
System.out.println(a);
System.out.print("short\t<=  long :\t");
b=(short)d;
System.out.println(b);
System.out.print("int\t<=  long :\t");
c=(int)d;
System.out.println(c);
System.out.print("char\t<=  long :\t");
g=(char)d;
System.out.println(g);

System.out.println("");
System.out.println("");

c=255;
System.out.print("byte\t<=  int :\t");
a=(byte)c;
System.out.println(a);
System.out.print("short\t<=  int :\t");
b=(short)c;
System.out.println(b);
System.out.print("char\t<=  int :\t");
g=(char)c;
System.out.println(g);

System.out.println("");
System.out.println("");

b=66;
System.out.print("byte\t<=  short :\t");
a=(byte)b;
System.out.println(a);
System.out.print("char\t<=  short :\t");
g=(char)b;
System.out.println(g);

System.out.println("");
System.out.println("");
```



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```
                g='a';
                System.out.print("byte\t<=  char :\t");
                a=(byte)g;
                System.out.println(a);
                System.out.print("short\t<=  char :\t");
                b=(short)g;
                System.out.println(b);
        }
}
```

Output

```
E:\RK_P\java\tutorials\t_2>javac Pro2.java

E:\RK_P\java\tutorials\t_2>java Pro2

        Implicit type casting(widening)
================================================
===============
short   <=  byte :      23
int     <=  byte :      23
long    <=  byte :      23
float   <=  byte :      23.0
double  <=  byte :      23.0


int     <=  short :     76
long    <=  short :     76
float   <=  short :     76.0
double  <=  short :     76.0


ASCII VALUE
int     <=  char :      65
long    <=  char :      65
float   <=  char :      65.0
double  <=  char :      65.0


long    <=  int :       567
float   <=  int :       567.0
double  <=  int :       567.0


float   <=  long :      9876.0
double  <=  long :      9876.0


double  <=  float :     89565.9921875
```

```
        Explicit type casting(narrowing)
================================================
===============
byte    <= double :     -108
short   <=  double :    9876
int     <=  double :    9876
long    <=  double :    9876
float   <=  double :    9876.0
char    <=  double :    ?


byte    <=  float :     -97
short   <=  float :     -31073
int     <=  float :     99999
long    <=  float:      99999
char    <=  float :     ?


byte    <=  long :      85
short   <=  long :      85
int     <=  long :      85
char    <=  long :      U


byte    <=  int :       -1
short   <=  int :       255
char    <=  int :       ÿ


byte    <=  short :     66
char    <=  short :     B


byte    <=  char :      97
short   <=  char :      97
```

Q-3

Write a program to demonstrate **integer overflow** — add 1 to `Integer.MAX_VALUE` and observe the result. Repeat for `byte` and `short`.

Code

```
class Pro3
{
        public static void main(String[] args)
        {
/*              int min=Integer.MIN_VALUE,of_m=min-1;
                System.out.println("Min value: "+min+"\nOverflow :"+of_m);*/
```



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```
                //in int automaticaly shift to (Min) value
                int max=Integer.MAX_VALUE,of=max+1;
                System.out.println("Max value: "+max+"\nOverflow :"+of);

        //for byte and short explicit type casting require because it become a int

                byte max_byte=Byte.MAX_VALUE,of_byte=(byte)(max_byte+1);
                System.out.println("Max value: "+max_byte+"\nOverflow
:"+of_byte);

                short max_short =
short.MAX_VALUE,of_short=(short)(max_short+1);
                System.out.println("Max value: "+max_short+"\nOverflow
:"+of_short);
        }
}
```

Output

```
E:\RK_P\java\tutorials\t_2>javac Pro3.java

E:\RK_P\java\tutorials\t_2>java Pro3
Max value: 2147483647
Overflow :-2147483648
Max value: 127
Overflow :-128
Max value: 32767
Overflow :-32768
```

Q-4

Write a program to accept a `char` and print its ASCII (Unicode) value, and vice versa (accept an int and print the corresponding character).

Code

```
class Pro4
{
        public static void main(String[] args)
        {
                char ch='A';
                int a=ch;
                System.out.println("Char ASCII value is : "+a);
                a=97;
                ch=(char)a;
                System.out.println("Int value to char is : "+ch);
        }
}
```



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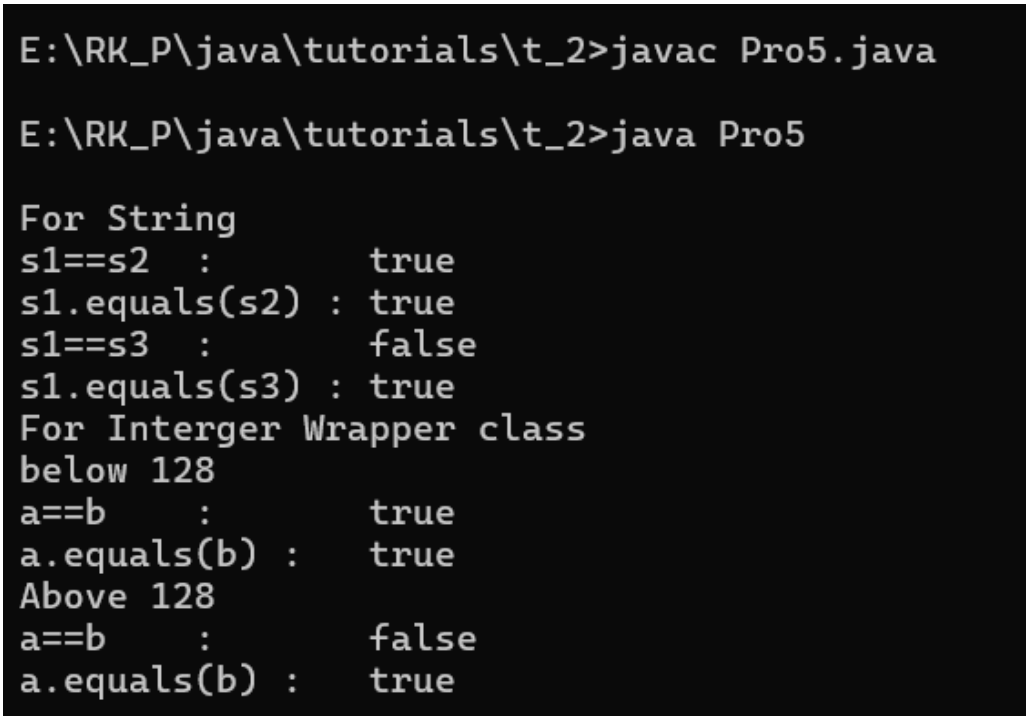
Output	`E:\RK_P\java\tutorials\t_2>javac Pro4.java`
	
	`E:\RK_P\java\tutorials\t_2>java Pro4`
	`Char ASCII value is : 65`
	`Int value to char is : a`

Q-5	Write a program to demonstrate the difference between `==` and `.equals()` for `String` and wrapper class objects (`Integer`, comparing values above and below 128 to show caching behavior).

Code	class Pro5
	{
	public static void main(String[] args)
	{
	System.out.println("");
	System.out.println("For String ");
	String s1="abc";
	String s2="abc";
	String s3=new String("abc");
	System.out.println("s1==s2\t:\t"+(s1==s2));
	System.out.println("s1.equals(s2) :\t"+s1.equals(s2));
	System.out.println("s1==s3\t:\t"+(s1==s3));
	System.out.println("s1.equals(s3) :\t"+s1.equals(s3));
	System.out.println("For Interger Wrapper class");
	Integer a=127;
	Integer b=127;
	System.out.println("below 128");
	System.out.println("a==b\t:\t"+(a==b));
	System.out.println("a.equals(b) :\t"+a.equals(b));
	a=129;
	b=129;
	System.out.println("Above 128");
	System.out.println("a==b\t:\t"+(a==b));
	System.out.println("a.equals(b) :\t"+a.equals(b));
	}
	}



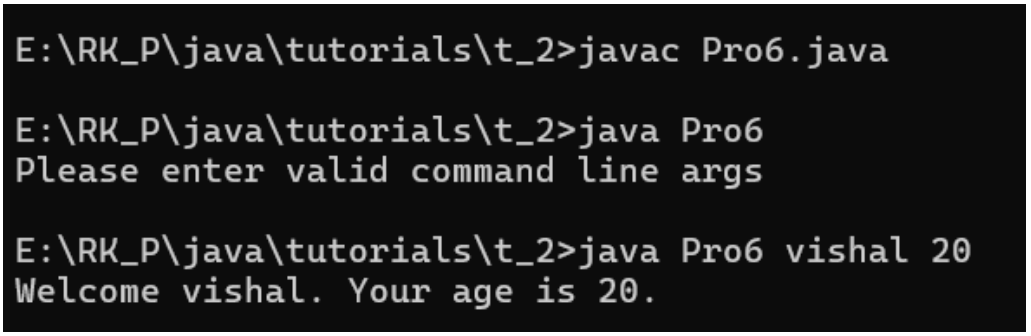
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Output	
Q-6	Write a program that accepts your name and age as command line arguments and prints a greeting message. Handle the case where arguments are missing.

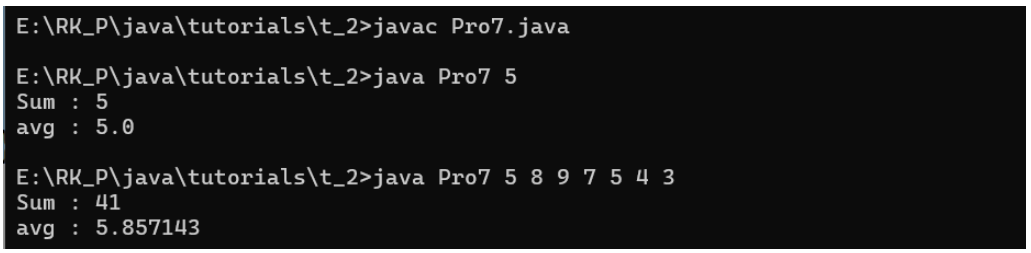
Code	class Pro6
	{
	public static void main(String[] args)
	{
	if(args.length==2)
	{
	System.out.println("Welcome "+args[0]+". Your age is "+args[1]+".");
	}
	else
	{
	System.out.println("Please enter valid command line args");
	}
	}
	}



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Output	
Q-7	Write a program to accept **n integers** as command line arguments, convert them from `String` to `int`, and print their sum and average.

Code	```
	class Pro7
	{
	public static void main(String[] args)
	{
	int sum=0;
	float avg=0;
	if(args.length>0)
	{
	for(int i=0;i<args.length;i++)
	{
	sum+=Integer.parseInt(args[i]);
	}
	
	avg=sum/(float)args.length;
	System.out.println("Sum : "+sum+"\navg : "+avg);
	}
	}
	}
	``

Output	
Q-8	Write a program to accept two numbers and an operator symbol (`+`, `-`, `*`, `/`) as command line arguments and perform a **simple calculator** operation. Handle divide-by-zero and invalid operator.



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Code

```
class Pro8
{
	public static void main(String[] args)
	{
		if(args.length==3)
		{
			int n1=Integer.parseInt(args[0]),n2=Integer.parseInt(args[1]);

			float res=0;
			char ch=args[2].charAt(0);
			if(n2==0 && ch=='/')
			{
				System.out.println("can not divide by zero");
			}
			else if(ch=='+' || ch=='-' || ch=='*' || ch=='/' )
			{
				switch(ch)
				{
					case '+':
						res=n1+n2;
					break;
					case '-':
						res=n1-n2;
					break;
					case '*':
						res=n1*n2;
					break;
					case '/':
						res=n1/n2;
					break;
				}
				System.out.println("Calculate : "+res);
			}
			else
			{
				System.out.println("Enter valid CLI ");
			}
		}
	}
}
```



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Output	E:\RK_P\java\tutorials\t_2>javac Pro8.java E:\RK_P\java\tutorials\t_2>java Pro8 E:\RK_P\java\tutorials\t_2>java Pro8 10 20 $ Enter valid CLI E:\RK_P\java\tutorials\t_2>java Pro8 10 20 + Calculate : 30.0
Q-9	Write a program to accept a sentence as multiple command line arguments (each word as a separate argument) and reconstruct/print the full sentence.

Code	class Pro9
	{
	public static void main(String[] args)
	{
	if(args.length>0)
	{
	for(int i=0;i<args.length;i++)
	{
	System.out.print(args[i]+" ");
	}
	}
	}
	}

Output	E:\RK_P\java\tutorials\t_2>javac Pro9.java E:\RK_P\java\tutorials\t_2>java Pro9 1 2 1 2 E:\RK_P\java\tutorials\t_2>java Pro9 1 2 5 6 8 1 2 5 6 8 E:\RK_P\java\tutorials\t_2>java Pro9 1 2 5 6 8 9 0 6 1 2 5 6 8 9 0 6
Q-10	Write a program to accept command line arguments and sort them — detect automatically whether they are numeric or alphabetic, and sort accordingly (use `try-catch` with `NumberFormatException` to decide).



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Code	class Pro10
	{
	public static void main(String[] args)
	{
	double[] num=new double[args.length];
	String[] alpha=new String[args.length];
	int n=0,a=0;
	if(args.length>0)
	{
	for(String arg:args)
	{
	try
	{
	Double d=Double.parseDouble(arg);
	num[n]=d;
	n++;
	}
	catch(NumberFormatException ex)
	{
	alpha[a]=arg;
	a++;
	}
	}
	for(int i=0;i<n;i++)
	System.out.print(num[i]+"\t");
	System.out.println();
	for(int i=0;i<a;i++)
	System.out.print(alpha[i]+"\t");
	}
	}
	}

Output	E:\RK_P\java\tutorials\t_2>javac Pro10.java E:\RK_P\java\tutorials\t_2>java Pro10 6 hi 1 helo 4 nj 5 6 nj kj j j j j k 9 6.0 1.0 4.0 5.0 6.0 9.0 hi helo nj nj kj j j j j k
Q-11	Write a program using `Scanner` to accept a student's name, roll number, and marks in 3 subjects, then calculate total, percentage, and grade.
Code	import java.util.Scanner; class Pro11 { public static void main(String[] args) { Scanner sc=new Scanner(System.in);



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```
String name,grade="";
int rollno;
float total=0,per;
int[] mark=new int[3];

System.out.print("Enter Name : ");
name=sc.nextLine();

System.out.print("Enter Roll No. :");
rollno=sc.nextInt();

for(int i=0;i<3;i++)
{
        System.out.print("Enter Mark of Subject "+(i+1)+" :");
        mark[i]=sc.nextInt();
        total+=mark[i];
        if(mark[i]<33)
        {
                grade="fail";
        }
}
per=total/3.00f;
if(grade=="fail")
{
}
else if(per>=91 && per<=100)
{
        grade="A1";
}
else if(per>=81 && per<=90)
{
        grade="A2";
}
else if(per>=71 && per<=80)
{
        grade="B1";
}
else if(per>=61 && per<=70)
{
        grade="B2";
}
else if(per>=51 && per<=60)
{
        grade="C1";
}
else if(per>=41 && per<=50)
{
        grade="C2";
```



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	}
	else if(per>=33 && per<=40)
	{
	grade="D";
	}
	else
	{
	grade="Fail";
	}
	System.out.println("Total : "+total);
	System.out.println("Percentage :"+per);
	System.out.println("Grade :"+grade);
	}
	}

Output	E:\RK_P\java\tutorials\t_2>java Pro11 Enter Name : vishal chotaliya Enter Roll No. :13 Enter Mark of Subject 1 :90 Enter Mark of Subject 2 :70 Enter Mark of Subject 3 :80 Total : 240.0 Percentage :80.0 Grade :B1
Q-12	Write a program using `Scanner` to accept multiple lines of input until the user types "exit" (use `nextLine()` in a loop), and count total words entered.

Code	import java.util.Scanner;
	class Pro12
	{
	public static void main(String[] args)
	{
	Scanner sc=new Scanner(System.in);
	int cnt=0;
	String s1="";
	System.out.println("Enter words if you want to exit enter 'Exit' ");
	while(!s1.equals("exit"))
	{
	s1=sc.nextLine();
	cnt++;



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```
                }
                System.out.println(cnt+" word you type. ");
        }
}
```

Output

```
E:\RK_P\java\tutorials\t_2>javac Pro12.java

E:\RK_P\java\tutorials\t_2>java Pro12
Enter words if you want to exit enter 'Exit'
hello
my name is vishal
my surname is chotaliya
my roll no is 13
ok then
i want by by
Exit
7 word you type.
```

Q-13

Write a program using `Scanner` to accept an integer `n`, then read `n` integers one by one into an array, and find the **maximum, minimum, and sum**.

Code

```
import java.util.Scanner;
class Pro13
{
        public static void main(String[] args)
        {
                Scanner sc=new Scanner(System.in);
                System.out.println("Enter how many number you want to type :");
                int n=sc.nextInt();
                int[] num=new int[n];
                for(int i=0;i<n;i++)
                {
                        System.out.print("Enetr number "+(i+1)+" :");
                        num[i]=sc.nextInt();
                }

                int max=num[0],min=num[0],sum=0;
                for(int i:num)
                {
                        if(i>max)
                                max=i;
```




```
                        if(i<min)
                                min=i;

                        sum+=i;
                }
                System.out.println("maximum number is : "+max);
                System.out.println("minimum number is : "+min);
                System.out.println("sum is : "+sum);
        }
}
```

Output

```
E:\RK_P\java\tutorials\t_2>javac Pro13.java

E:\RK_P\java\tutorials\t_2>java Pro13
Enter how many number you want to type :
5
Enetr number 1 :12
Enetr number 2 :56
Enetr number 3 :23
Enetr number 4 :45
Enetr number 5 :20
maximum number is : 56
minimum number is : 12
sum is : 156
```

Q-14

Write a program using `Scanner` to build a simple **menu-driven application** (e.g., Add/View/Delete a record) that runs in a loop until the user selects "Exit" — this simulates a real-world console application.

Code

```
import java.util.*;
class Pro14
{
        public static void main(String[] args)
        {
                Scanner sc=new Scanner(System.in);
                ArrayList<String> records=new ArrayList<>();
                Boolean isRun=true;
                while(isRun)
                {
                        System.out.println("==========Welcome to Record
App==========");
                        System.out.println("Press 1 :  Add records");
```



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```
                        System.out.println("Press 2 :  View records");
                        System.out.println("Press 3 :  Delete records");
                        System.out.println("Press 4 :  Exit from the App");

System.out.println("================================================
==========");
                        System.out.print("Enter Your Choice : ");
                        int ch=sc.nextInt();
                        System.out.println();
                        switch(ch)
                        {
                                case 1:
                                        System.out.print("How many record you
want to enter : ");
                                        int r1=sc.nextInt();

System.out.println("--------------------------------------------------------------------------");
                                        System.out.println("Entre records : ");

System.out.println("--------------------------------------------------------------------------");
                                        sc.nextLine();
                                        for(int i=1         ;i<=r1;i++)
                                        {
                                                String temp=sc.nextLine();
                                                records.add(temp);
                                        }
                                        System.out.println("Your records added
successfully!");
                                        System.out.println();
                                break;
                                case 2:

System.out.println("--------------------------------------------------------------------------");
                                        System.out.println("Your records : ");

System.out.println("--------------------------------------------------------------------------");
                                        for(String s1:records)
                                                System.out.println(s1);
                                break;
                                case 3:
                                        for(int i=0;i<records.size();i++)
                                        {
                                                System.out.println("At index
"+(i+1)+" "+records.get(i));
                                        }
                                        System.out.print("Enter index you want to
delete record :");
                                        int ind=sc.nextInt();
```



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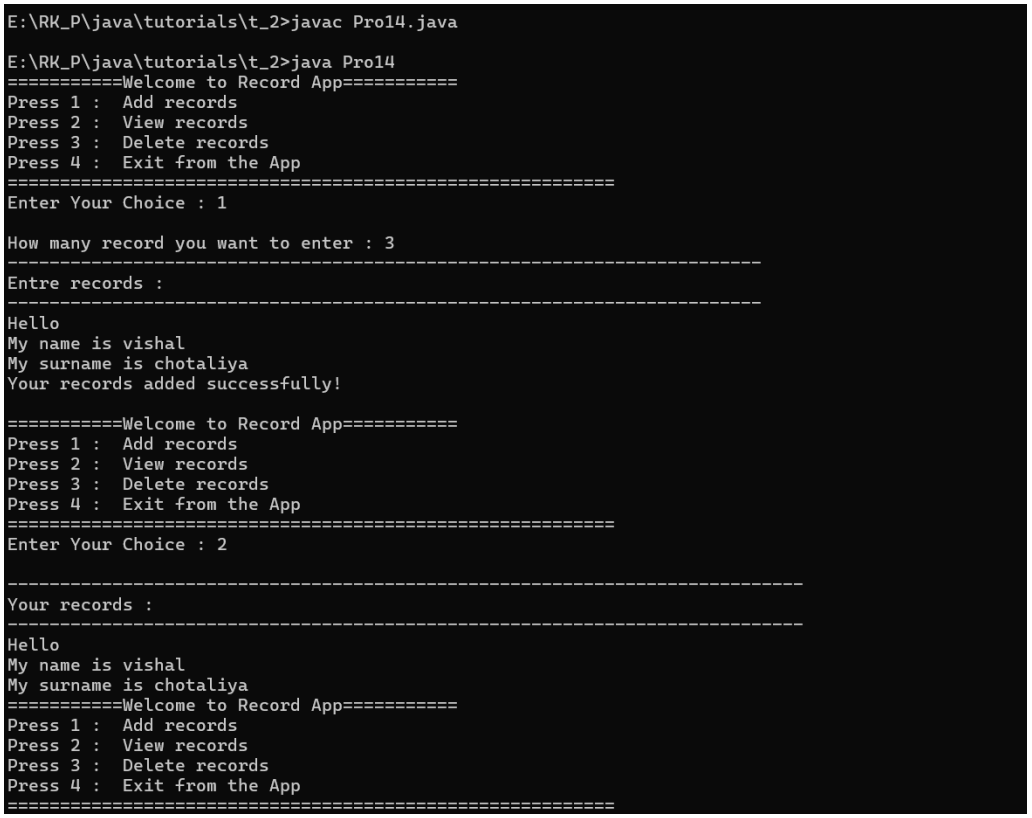
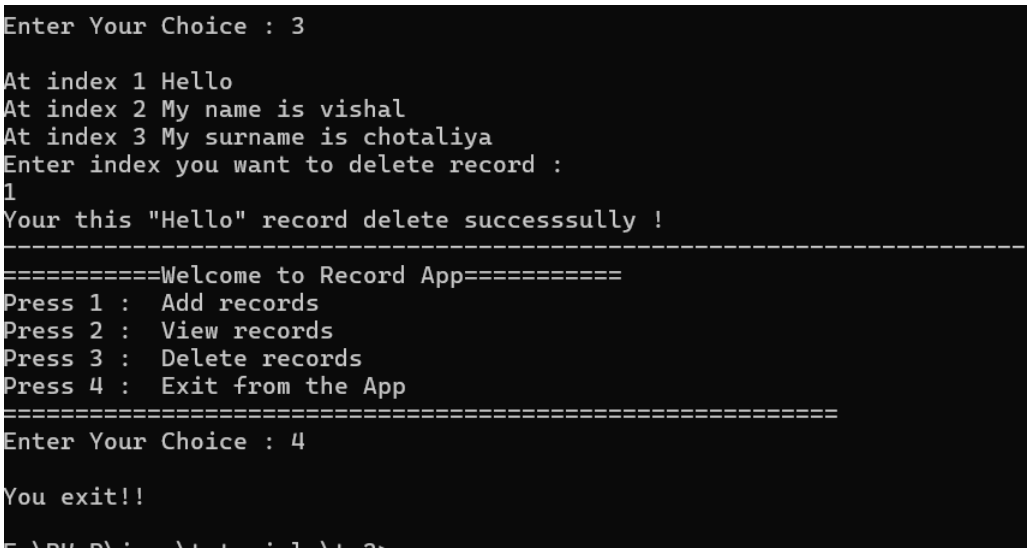
```
                                        try
                                        {
                                                String
remove=records.remove(ind-1);
                                                System.out.println("Your this
\""+remove+"\" record delete successsully !");
                                        }
                                        catch(IndexOutOfBoundsException ex)
                                        {
                                                System.out.println("Entre valid
choice your enter index is not found.");
                                        }

System.out.println("--------------------------------------------------------------------------");
                                break;
                                case 4:
                                        isRun=false;
                                        System.out.println("You exit!!");
                                break;
                                default:
                                        System.out.println("Enter your choice from
only(1-4)");
                        }

                }
        }
}
```



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Output	 
Q-15	Write a **combined program**: accept the number of employees as a **command line argument**, then use `Scanner` to input each employee's name (`String`), ID (`int`), and salary (`double`). Store them, apply a 10% bonus to salaries above a threshold, and display a formatted



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	report. This combines all three concepts.

Code	import java.util.Scanner;
	class Pro15
	{
	public static void main(String[] args)
	{
	int numOfEmployee=0;
	double threshold=30000;
	if(args.length==1)
	numOfEmployee=Integer.parseInt(args[0]);
	
	Scanner sc=new Scanner(System.in);
	String[] ename=new String[numOfEmployee];
	int[] eid=new int[numOfEmployee];
	double[] esalary=new double[numOfEmployee];
	for(int i=0;i<numOfEmployee;i++)
	{
	
	System.out.print("Enter "+(i+1)+" Employee name : ");
	
	ename[i]=sc.nextLine();
	System.out.print("Enter "+(i+1)+" Employee Id : ");
	eid[i]=sc.nextInt();
	System.out.print("Enter "+(i+1)+" Employee Salary: ");
	esalary[i]=sc.nextDouble();
	sc.nextLine();
	if(esalary[i]>threshold)
	esalary[i]+=(esalary[i]*0.1);
	}
	
	System.out.println("----------");
	System.out.println("Employee Details : ");
	
	System.out.println("----------");
	System.out.println("Employee Id\t||\tEmployee Name\t||\tEmployee salary");
	System.out.println();
	for(int i=0;i<numOfEmployee;i++)
	{
	
	System.out.println(eid[i]+"\t\t||\t"+ename[i]+"\t\t||\t"+esalary[i]);
	}
	}
	}



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Output	E:\RK_P\java\tutorials\t_2>javac Pro15.java E:\RK_P\java\tutorials\t_2>java Pro15 3 Enter 1 Employee name : vishal Enter 1 Employee Id : 101 Enter 1 Employee Salary: 90000 Enter 2 Employee name : khushal Enter 2 Employee Id : 102 Enter 2 Employee Salary: 98000 Enter 3 Employee name : raj Enter 3 Employee Id : 103 Enter 3 Employee Salary: 99000 --- Employee Details : --- Employee Id || Employee Name || Employee salary 101 || vishal || 99000.0 102 || khushal || 107800.0 103 || raj || 108900.0 E:\RK_P\java\tutorials\t_2>java Pro15 Enter valid CLI